Qixin Xu

Email ◇ Google Scholar ◇ Github

EDUCATION

Tsinghua University 2023 - 2027(expected)

B.S. in Computer Science; GPA: 3.9/4.0; Rank: 9/172

- **Selected A^+/A Courses**: Linear Algebra, Probability and Statistics, Introduction to Artificial Intelligence, Introduction to Statistical Learning, Deep Learning, Multi-Modal Machine Learning, Programming and Training, Data Mining, Writing and Communication, English for Academic Purposes: Research Paper Writing etc.
- **Academic Interests**: Reinforcement Learning, LLM (Agents)

RESEARCH PUBLICATION

** EQUAL CONTRIBUTIONS, † CORRESPONDING AUTHOR*

(1) **Qixin Xu***, Haozhe Wang*, Che Liu, Fangzhen Lin, Wenhu Chen†. **DocThinker: Towards Unified thinking in Documents**, *Under peer review*, 2026.

(2) **Qixin Xu**, Bohan Lyu, Zihan Zhu, Yesai Wu, Zhong Zhang, Haotian Chen, Xin Cong, Xiaojiang Liu, Zhiyuan Liu†, Maosong Sun. **Diversity-aware Training for Test-time Scaling**, *Under peer review*, 2026.

(3) Haoxiang Zhang*, **Qixin Xu***, Zhuofeng Li, Lei Zhang, Pengcheng Jiang, Yu Zhang†, Julian McAuley†. **Masking Stale Observations Helps Search Agents–Until It Doesn't: A Regime Map and Its Mechanism**, *Under peer review*, 2026

(4) Haozhe Wang, **Qixin Xu**, Changpeng Wang, Taofeng Xue, Chong Peng, Wenhu Chen†, Fangzhen Lin†. **Bad Seeing or Bad Thinking? Rewarding Perception-Reasoning Synergy for Scalable Vision-Language Reasoning**, *ICML 2026 (Oral)*

(5) Bohan Lyu*, Yucheng Yang*, Siqiao Huang*, Jiaru Zhang*, **Qixin Xu***, Xinghan Li*, Xinyang Han*, Yicheng Zhang*, Huaqing Zhang*, Runhan Huang, Kaicheng Yang, Zitao Chen, Wentao Guo, Junlin Yang, Xinyue Ai, Wenhao Chai, Yadi Cao, Ziran Yang, Kun Wang, Dapeng Jiang, Huan-ang Gao, Shange Tang, Chengshuai Shi, Simon S Du, Max Simchowitz, Jiantao Jiao, Dawn Song, Chi Jin†. **MLS-Bench: A Holistic and Rigorous Assessment of AI Systems on Building Better AI**, *Under peer review*, 2026

(6) Ganqu Cui*, Lifan Yuan*, Zefan Wang*, Hanbin Wang*, Wendi Li*, Bingxiang He*, Yuchen Fan*, **Qixin Xu***, Tianyu Yu*, Weize Chen, Jiarui Yuan, Huayu Chen, Kaiyan Zhang, Xingtai Lv, Shuo Wang, Yuan Yao, Xu Han, Hao Peng, Yu Cheng, Zhiyuan Liu, Maosong Sun, Bowen Zhou, Ning Ding†. **Process reinforcement through implicit rewards**, *TMLR 2026*

(7) Haozhe Wang, **Qixin Xu**, Che Liu, Junhong Wu, Fangzhen Lin†, Wenhu Chen†. **Emergent hierarchical reasoning in llms through reinforcement learning**, *ICLR 2026*

(8) Haozhe Wang, Haoran Que, **Qixin Xu**, Minghao Liu, Wangchunshu Zhou, Jiazhan Feng, Wanjun Zhong, Wei Ye, Tong Yang, Wenhao Huang, Ge Zhang†, Fangzhen Lin†. **Reverse-Engineered Reasoning for Open-Ended Generation**, *ICLR 2026*

(9) Tianyu Yu, Haoye Zhang, Qiming Li, **Qixin Xu**, Yuan Yao†, Da Chen, Xiaoman Lu, Ganqu Cui, Yunkai Dang, Taiwen He, Xiaocheng Feng, Jun Song, Bo Zheng, Zhiyuan Liu†, Tat-Seng Chua, Maosong Sun. **RLAIF-V: Open-Source AI Feedback Leads to Super GPT-4V Trustworthiness**, *CVPR 2025 (Highlight)*

(10) Pan Lu*, Yifan Gao*, William G Peng, Haoxiang Zhang, Kunlun Zhu, Elektra K Robinson, **Qixin Xu**, Masakazu Kotaka, Harrison G Zhang, Bingxuan Li, Anthony L Shiver, Yejin Choi, Kerwyn Casey Huang, Justin L Sonnenburg†, James Zou†. **Eubiota: Modular Agentic AI for Autonomous Discovery in the Gut Microbiome**, *Submission to Nature*

RESEARCH EXPERIENCE

Stanford NLP Group November 2025 – Present

Research Intern(advised by Prof. James Zou)

- **Multi-Agent Systems for Scientific Discovery via Reinforcement Learning:** Developing a multi-agent Reinforcement Learning framework and infrastructure for autonomous scientific discovery; achieved State-of-the-Art performance on domain-specific benchmarks and validated findings through downstream human evaluation.

TIGER AI Lab June 2025 – Present

Research Intern(advised by Prof. Wenhu Chen)

- **Unlock the Implicit Mechanism of Reinforcement Learning:** Acknowledged two-stage training dynamics in RL, providing interpretability for black-box phenomena like "Aha moments" and "Length Scaling Effects". Proposed HIerarchy-Aware Credit Assignment, outperforming GRPO across diverse model families. Paper was accpeted by *ICLR 2026*.
- **Multimodal reasoning under stress tests and its mechanisms:** Developed coarse-to-fine algorithms and curated high-quality multimodal datasets, achieving state-of-the-art results on document reasoning benchmarks. Demystified the coupling between visual perception and text reasoning, improving general-purpose visual reasoning. Paper was accpeted by *ICML 2026 (Spotlight)*.

Tsinghua NLP Lab July 2024 – November 2025

Research Intern (advised by Prof. Zhiyuan Liu)

- **Enhancing Diversity in Large Language Models:** Identified that SFT and RL pipelines contribute to reduced diversity (mode collapse) in generation space; developed a diversity-aware data pipeline and integrated noisy embeddings as an implicit diversity signal, improving pass@k and other diversity metrics.
- **Enhancing Visual Reasoning in Multimodal Language Models:** Explored tool-use RL for visually intensive tasks (counting, grounding); developed a Tool-Use RL framework in veRL; improved Qwen2.5-VL-7B-Instruct on key counting/grounding benchmarks.
- **PRIME: Implicit Reward Modeling for Reasoning:** Proposed PRIME (**P**rocess **R**einforcement through **IM**plicit r**E**wards) enabling online PRM updates using only policy rollouts and outcome labels via implicit process rewards; studied the role of dense rewards in RL. Paper was accepted by *TMLR 2026*.
- **Multi-modal Alignment through AI Feedback:** Investigated feedback from open-source models to reduce VLM hallucination; used a DPO-trained model as reward model for inference-time scaling. Paper was accepted by *CVPR 2025 (Hightlight)*.

PROJECT EXPERIENCE

Contributor of MiniCPM-V 2.6, MiniCPM-o 2.6 and MiniCPM-V 4.5 July 2024 - July 2025

Worked with OpenBMB Team

- Develop a simple yet effective pipeline to reduce MiniCPM-V 2.6 and MiniCPM-o 2.6's hallucination
- Design particular format of pre-training data to enable the training of streaming ability of MiniCPM-o 2.6
- Technical report was accepted by *CVPR 2026*

SKILLS

Languages: English(CET-6: 651/710 (Listening: 249/249)), Chinese.
Programming: Python, C/C++, System Verilog, LATEX, etc.
Tools: vllm, Pytorch, DeepSpeed, Git, Shell, Vim, Django, ssh, gdb/pdb, etc.
Frameworks: veRL, Slime, EasyR1, LLaMA-Factory